GONZALO ANDRÉS VIDAL PEÑA

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EDUCATION

Newcastle University 2023
Computer Science PhD *Interdisciplinary Computing and Complex BioSystems Research Group*
Thesis: Automated design, build, test, learn workflows to engineer synthetic genetic networks

Pontifical Catholic University of Chile 2019
Biochemistry Bachelor and Professional Title *Honors*

University of Chile 2012
Natural and Exact Sciences Bachelor *Honors*

WORK EXPERIENCE

University of Colorado Boulder 2023 - Present
Postdoctoral Researcher

Developing and architecting data management workflows for synthetic biology experimental data and metadata. Deploying distributed and robust databases, modular software for the Design, Build, Test, Learn (DBTL) cycle, and solutions for data collection. Project management and reporting for the CU collaboration to Engineering Biology for Underwater and Ground Sensors (EBUGS) and the Center for Harnessing Microbiota from Military Environments (CHARMME). Mentored by Prof. Chris Myers.

SBOL Industrial, BioDesign Automation Consortium (BDAC) 2022
Research Intern

Developing a high-level language for creating genetic designs using the Synthetic Biology Open Language (SBOL). Developing a workflow to go from plasmid design in SBOL to assembly instructions in the OT-2 liquid handling robot connecting the Design and Build stages. Mentored by Dr. Jacob Beal.

Institute for Biological and Medical Engineering 2019
Automation and Robotics Engineer

DNA assembly automation. Developing automated protocols to perform Golden Gate, Gibson and BASIC assembly using liquid handling robots. Developing automated pipelines for genetic network characterization. Providing DNA assembly service. Mentored by Dr. Timothy Rudge.

Monsanto/ Bayer 2017 - 2018
Research Intern

Researching the use satellite and drone images for predicting plant water usage. Creation and management of a project to implement the predictions for water management on *Brassica*. Limitless innovation and entrepreneurship finalist from Pontifical Catholic University of Chile.

Milandu 2016 - 2019
CTO and Co-Founder

Startup in Maule valley, Chile. Water usage prediction, management and remote sensing for productive crops. Production and distribution of amaranth grains.

RESEARCH ACCOMPLISHMENTS

My research has advanced synthetic biology by building the conceptual, technical, computational, and community foundations needed to make the field more collaborative, predictable, reproducible, and scalable. I have contributed to core standards, created interoperable software ecosystems, and developed automated and data-driven workflows that bridge computation and experimentation. On my research I have contributed to the development of many software tools: LOICA, PUDU, Tricahue, Flapjack, MiNuD, SynBioSuite, SeqTrainer, Excel2SBOL, SBOL-utilities, CellModeller, and more.

AWARDS

Best Postdoc Mentor, *ECEE University of Colorado Boulder.*	2025
Best Poster Award, *International Workshop on BioDesign Automation.*	2022
School of Computing scholarship, *Newcastle University.*	2021
Biological and Medical Engineering scholarship, *Pontifical Catholic University of Chile.*	2019
Academic Excellence, *Pontifical Catholic University of Chile.*	2018
University of Chile scholarship, *University of Chile.*	2011
Bicentenary scholarship, *Ministry of Education, Chile.*	2011

FUNDRAISING

Grant writing, *NSF Systems and Synthetic Biology; lead PI Chris Myers, ongoing.*	2025
Grant awarded, SeqTrainer: ML package for Synthetic Biology, *Google Open Source.*	2025
Grant awarded, SBOL2-SBOL3 converter, *Google Open Source.*	2025
Grant awarded, Automated synthesability scores for SBOL, *Google Open Source.*	2024
Grant writing, *UKRI/BBSRC-NSF/BIO; lead PIs Tim Rudge & Chris Myers, rejected.*	2022
Grant writing, *UKRI/BBSRC Synthetic Biology; lead PI Tim Rudge, rejected.*	2021
Grant awarded, Laboratory 4.0, *DAE Pontifical Catholic University of Chile.*	2020
Grant awarded, Plant growing automation, *DAE Pontifical Catholic University of Chile.*	2020
Grant writing, *ANID Chile; lead PI Tim Rudge, awarded.*	2020
Grant awarded, BME student association, *DAE Pontifical Catholic University of Chile.*	2019
Grant writing, *ANID Chile; lead PI Tim Rudge, rejected.*	2019
Startup seed capital awarded, *CORFO Chilean Economic Development Agency.*	2016

SERVICE

Host, NonaTalks — 2025

NonaTalks brings together innovators, experts, and enthusiasts for a compelling showcase of progress and potential in open-source synthetic biology software. From innovative software tools to creative design approaches, these presentations highlight the ingenuity and hard work of the participants pushing the boundaries of synthetic biology.

Chair, International Workshop on BioDesign Automation (IWBDA) — 2023-2024

IWBDA is an annual conference that provides a forum for cross-disciplinary discussion, with the aim of seeding and fostering collaboration between the researchers from the synthetic biology, systems biology, and design automation communities. Served as Workshop Chair in 2023 and Program Chair in 2024.

Chair, SynBioNet NEUK — 2022-2023

Synthetic Biology Networking at the North East of UK, led by Newcastle University and Northumbria University. Organization of meetings and presentations on synthetic biology research that foster collaboration between academic, industry and government actors.

Member, iGEM Engineering Committee — 2020-present

Active member on the general, software and interlab committees. Automation interlab leader, developing, planning and conducting inter laboratory studies using automation.

Editor, SBOL 2020-2024

Editor of the Synthetic Biology Open Language specification. Lead, develop, maintain and coordinate community software, activities and events. Continue contributing to the steering committee and developers of the SBOL ecosystem.

Postgraduate Student Representative, Biological and Medical Engineering 2019-2021

Creation, funding acquisition, execution and leading of interdisciplinary projects. Representation of BME postgraduate students in the school.

TEACHING EXPERIENCE

Invited Lectures

Engineering Genetic Circuits, *University of Colorado Boulder* 2025
Advanced Synthetic Biology, *Newcastle University.* 2022
Introduction to Synthetic Biology, *Newcastle University.* 2022
Synthetic Biology, *Pontifical University Catholic of Chile.* 2018, 2019
Complex Systems, *Pontifical University Catholic of Chile.* 2018, 2019

Teaching Assistant Graduate Courses

Data Visualization, *Newcastle University.* 2023,2024
Advanced Synthetic Biology, *Newcastle University.* 2022
Introduction to Synthetic Biology, *Newcastle University.* 2022

Teaching Assistant Undergraduate Courses

Software Engineering, *Newcastle University.* 2023
Contemporary Topics in Computing, *Newcastle University.* 2023
Computer Systems Design and Architectures, *Newcastle University.* 2023
Biomedical Data Analytics and AI, *Newcastle University.* 2022
Complex Systems, *Pontifical University Catholic of Chile.* 2018, 2019
Synthetic Biology, *Pontifical University Catholic of Chile.* 2017, 2018, 2019

MENTORSHIP

Mentees

Carolus Vitalis, AI for genetic annotation and design. *BME & IQ Biology PhD student, CU Boulder.*
Hatem Habdelraham, synthetic biology approach for stem cell design. *BME PhD student, CU Boulder.*
Chunxiao Liao, AI for reproducibility and agentic scientific discovery. *CS PhD student, CU Boulder.*
Daniel Fang, federated and robust database for synthetic biology. *CS PhD student, CU Boulder.*
William Mo, biological and computational tools for cyber-biosecurity. *BME PhD student, CU Boulder.*
Zane Perry, ML approach for predicting gene expression. *Applied Math MSc student, CU Boulder.*
Saanika Fadia, standardized metadata collection and analysis. *CS undergrad student, CU Boulder.*
Oscar Rodriguez, lab automation hyper-parameter optimization. *CS undergrad student, CU Boulder.*
Kenzo Schwab, graphical user interfaces for synthetic biology. *CS undergrad student, CU Boulder.*
Sai Wong, SeqTrainer: ML package for synthetic biology. *intern, Google.*
Harsh Sharma, SBOL2-SBOL3 converter; *intern, Google.*
Ryan Greer, software tool for assembly planning. *CS undergrad student, CU Boulder.*
Kerem Gurkan, software tool for experimental design. *CS undergrad student, CU Boulder.*
Peter Hindes, SynBioHub documentation, CI/CD and DevOps. *ECEE undergrad student, CU Boulder.*
Luke Dysart, connecting DNA design to automated build. *ECEE undergrad student, CU Boulder.*
Parker Ackerknecht, automating synthetic biology protocols. *ECEE undergrad student, CU Boulder.*
Derick Sayavong, DNA design and simulation platforms. *CS undergrad student, CU Boulder.*

Taiisia Shertsikova, automating data collection. *CS undergrad student, CU Boulder.*
William Dravenstott, plate reader experimental database. *CS undergrad student, CU Boulder.*
Dylan Smith, plate reader experimental database. *CS undergrad student, CU Boulder.*
Samuel Ball, automated data collection from plate reader outputs. *CS undergrad student, CU Boulder.*
Rosalie Hughes, experimental data design UI. *Mechanical Engineering undergrad student, CSU.*
Sai Samineni, standardized experimental data collection. *BME PhD student, CU Boulder.*
Johan Guillén, automated synthesability scores for SBOL. *intern, Google.*
Luisa Konkina, implementing consortia distributed genetic networks. *CS PhD student, NU.*
Luisa Konkina, simulating consortia distributed genetic networks. *Synthetic Biology MSc student, NU.*
Carolus Vitalis, automated DNA assembly. *Molecular Biotechnology undergrad student, UChile.*
Ignacio Retamal, automated protocols for protein activity assays. *BME undergrad student, PUC.*
Sofia Figueroa, microscopy method to capture growing colonies. *BME undergrad student, PUC.*

Thesis committee

- Fernanda Lorca, GC-AI: A webtool for translating hand-writed genetic circuits to SBOL 3.1 standard. *MSc CS and Telecommunications; supervisor Martín Gutiérrez Pescarmona.*
- Catalina Lorca, SimBOL: A modular architecture for converting SBOL3 genetic circuit designs to multiple simulation. *MSc CS and Telecommunications; supervisor Martín Gutiérrez Pescarmona.*

TECHNICAL STRENGTHS

Drylab

Modeling and Analysis	ODE, Stochastic, AI, Complex Systems, ABM, Signal Analysis.
Programming Languages	Python, JavaScript, R, Matlab, SQL, GO, Julia.
Main Python Packages	ScyPy, NumPy, Pandas, Scikit-Image, DataStax.
AI Packages	TensorFlow, Keras, PyTorch, Scikit-Learn, LangChain, LangGraph.
Visualization Packages	Matplotlib, Seaborn, Plotly, LangFlow, Bokeh.
Other Software & Tools	ImageJ, GraphPad, Latex, MS Office, Affinity Designer.

Wetlab

Test Equipment	Plate reader, Flow cytometry, Microscopy, Sequencing.
Hosts/Chassis	Bacteria, Fungi, Plants, Mammalian cell lines and primary cultures.
Automation Equipment	OT-2, Echo Acoustic Liquid Handling, CyBio FeliX, PIXL.

Professional

Certifications	Leadership, Project management, HPC, Technology transfer.
Soft skills	Teamwork, Critical thinking, Empathy, Creativity.
Languages	English (Advanced), Spanish (Native).

PUBLICATIONS

Book Chapters

- Vidal, G., Vitalis, C., & Guillén, J. (2024). Standardized Golden Gate Assembly Metadata Representation Using SBOL. In Golden Gate Cloning: Methods and Protocols (pp. 89-104). New York, NY: Springer US.
https://doi.org/10.1007/978-1-0716-4220-7_6
- Vidal, G., Vitalis, C., Matúte, T., Núñez, I., Federici, F., & Rudge, T. J. (2024). Genetic Network Design Automation with LOICA. In Synthetic Biology: Methods and Protocols (pp. 393-412). New York, NY: Springer US.
https://doi.org/10.1007/978-1-0716-3658-9_22
- Vitalis, C., Feliú, G. Y., Vidal, G., Silva, M. M., Matúte, T., Núñez, I., ... & Rudge, T. J. (2024). Flapjack: Data Management and Analysis for Genetic Circuit Characterization. In Synthetic Biology:

Methods and Protocols (pp. 413-434). New York, NY: Springer US.
https://doi.org/10.1007/978-1-0716-3658-9_23

Journals

- Buecherl, L., Buson, F. X., Sørensen, G. H., Vitalis, C., Kubaczka, E., Vidal, G., ... & Vaidyanathan, P. (2025). A decade of SBOL Visual: growing adoption of a diagram standard for engineering biology. ACS Synthetic Biology.
https://doi.org/10.1021/acssynbio.5c00417
- Macuada, J., Molina-Riquelme, I., Vidal, G., Pérez-Bravo, N., Vásquez-Trincado, C., Aedo, G., ... & Eisner, V. (2024). OPA1 and disease-causing mutants perturb mitochondrial nucleoid distribution. Cell Death & Disease, 15(11), 870.
https://doi.org/10.1038/s41419-024-07165-9
- Beal, J., Selvarajah, V., Chambonnier, G., Haddock, T., Vignoni, A., Vidal, G., & Roehner, N. (2023). Standardized representation of parts and assembly for build planning. ACS synthetic biology, 12(12), 3646-3655.
https://doi.org/10.1021/acssynbio.3c00418
- Buecherl, L., Mitchell, T., Scott-Brown, J., Vaidyanathan, P., Vidal, G., Baig, H., ... & Myers, C. (2023). Synthetic biology open language (SBOL) version 3.1.0. Journal of Integrative Bioinformatics, 20(1), 20220058.
https://doi.org/10.1515/jib-2022-0058
- Aldulijan, I., Beal, J., Billerbeck, S., Bouffard, J., Chambonnier, G., Vidal, G., ... & Vignoni, A. (2023). Functional synthetic biology. Synthetic Biology, 8(1), ysad006.
https://doi.org/10.1093/synbio/ysad006
- Samineni, S. P.*, Vidal, G.*, Vitalis, C., Feliú, G. Y., Rudge, T. J., Myers, C. J., & Mante, J. (2023). Experimental data connector (XDC): integrating the capture of experimental data and metadata using standard formats and digital repositories. ACS Synthetic Biology, 12(4), 1364-1370.
https://doi.org/10.1021/acssynbio.2c00669
- Vidal, G., Vitalis, C., Muñoz Silva, M., Castillo-Passi, C., Yáñez Feliú, G., Federici, F., & Rudge, T. J. (2022). Accurate characterization of dynamic microbial gene expression and growth rate profiles. Synthetic Biology, 7(1), ysac020.
https://doi.org/10.1093/synbio/ysad006
- Vidal, G., Vitalis, C., & Rudge, T. J. (2022). LOICA: Integrating models with data for genetic network design automation. ACS Synthetic Biology, 11(5), 1984-1990.
https://doi.org/10.1021/acssynbio.1c00603
- Yanez Feliu, G., Earle Gomez, B., Codoceo Berrocal, V., Munoz Silva, M., Nuñez, I. N., Matute, T. F., Vidal, G. ... & Rudge, T. J. (2020). Flapjack: Data management and analysis for genetic circuit characterization. ACS Synthetic Biology, 10(1), 183-191.
https://doi.org/10.1021/acssynbio.0c00554
- Yáñez Feliú*, G., Vidal, G.*, Muñoz Silva, M., & Rudge, T. J. (2020). Novel tunable spatio-temporal patterns from a simple genetic oscillator circuit. Frontiers in Bioengineering and Biotechnology, 8, 893.
https://doi.org/10.3389/fbioe.2020.00893
- Torres, G., Morales, P. E., García-Miguel, M., Norambuena-Soto, I., Cartes-Saavedra, B., Vidal-Peña, G., ... & Chiong, M. (2016). Glucagon-like peptide-1 inhibits vascular smooth muscle cell dedifferentiation through mitochondrial dynamics regulation. Biochemical pharmacology, 104, 52-61.
https://doi.org/10.1016/j.bcp.2016.01.013

Preprints

- Rudge, T. J., & Vidal, G. (2022). Phase-based genetic logic circuits. BioRxiv, 2022-12. .
https://doi.org/10.1101/590299

- Muñoz Silva, M. A., Matute, T., Nuñez, I., Valdes, A., Ruiz, C. A., Vidal Peña, G. A., ... & Rudge, T. J. (2019). Phase space characterization for gene circuit design. BioRxiv, 590299. https://doi.org/10.1101/2022.12.13.520289

PRESENTATIONS

Talks

- SynBioSuite 2: Closing and connecting the synthetic biology workflow. NonaTalks. 2025
- Software tools for the Synthetic Biology DBTL cycle. International Genetically Engineered Machine (iGEM). 2022

Workshops

- Software tools for synthetic biology. Synthetic Biology: Engineering, Evolution & Design (SEED). 2023
- Software tools for synthetic biology. Synthetic Biology: Engineering, Evolution & Design (SEED). 2022
- Software tools for synthetic biology. Hackathon of the Computational Modelling in Biology Network (HARMONY). 2022
- Flapjack: Data management and analysis for genetic circuit characterization. International Workshop on Bio-Design Automation (IWBDA). 2021

Posters

- SeqTrainer: Encoding Synthetic Biology Data for Machine Learning. International Workshop on Bio-Design Automation (IWBDA). 2025.
- Excel-SBOL Converter Version 2. International Workshop on Bio-Design Automation (IWBDA). 2025.
- BuildPlanner: A tool for connecting the Design and Build stages of the DBTL cycle. International Workshop on Bio-Design Automation (IWBDA). 2025.
- An end-to-end standardized data management workflow for synthetic biology. Synthetic Biology: Engineering, Evolution & Design (SEED). 2025.
- SynBioSuite 2: Improving encoding of genetic designs and extending DBTL coverage. International Workshop on Bio-Design Automation (IWBDA). 2024.
- PUDU: Build and Test Automation for SynBio. Hackathon of the Computational Modelling in Biology Network (HARMONY). 2023.
- A proposal for connecting and automating the synthetic biology Design Build Test Learn cycle. Synthetic Biology UK (SBUK). 2022.
- A proposal for connecting and automating the Synthetic Biology Design Build Test Learn cycle. SAGE PGR Conference. 2022.
- *Best poster award* — Standardizing the representation of parts and devices for Build Planning. International Workshop on Bio-Design Automation (IWBDA). 2022.
- Steps towards functional synthetic biology. International Workshop on Bio-Design Automation (IWBDA). 2022.
- Experimental data converter. International Workshop on Bio-Design Automation (IWBDA). 2022.
- LOICA 1.2: Genetic network design automation for spatio-temporal patterns. Synthetic Biology: Engineering, Evolution & Design (SEED). 2022.
- LOICA 1.2: Genetic network design automation for spatio-temporal patterns. SNES FEST. 2022.
- LOICA: Logical operators for integrated cell algorithms. International Workshop on Bio-Design Automation (IWBDA). 2021.
- LOICA: Logical operators for integrated cell algorithms. The 1st International Biodesign Research Conference (IBDRC). 2020.
- LOICA: Logical operators for integrated cell algorithms. Computational Modeling in Biology Network (COMBINE). 2020.

- Self-organized Patterns from a Synthetic Genetic Oscillator in Bacterial Colonies. International Society for Microbial Ecology Latin America (ISME-LA). 2019.
- Open-source paper-fluidic device for bacterial culture, communication and biocomputation. International Society for Microbial Ecology Latin America (ISME-LA). 2019.
- Open-source paper-fluidic device for bacterial culture, communication and biocomputation. Synthetic Biology: Engineering, Evolution & Design (SEED). 2019.
- Modelling non-equilibrium polysome dynamics with totally asymmetric simple exclusion process (TASEP). ISCB-LA SOIBIO EMBNET. 2018.
- Control of phenotypic switching in vascular smooth muscle cells by mitochondrial metabolism. XXV Interamerican congress on cardiology and vascular surgery. 2015.
- Metabolic shift induced by PDGF-BB involves mitochondrial fragmentation in VSMCs. XXXVII Annual meeting of the chilean society of biochemistry and molecular biology. 2014.
- Mitochondrial metabolism and the control of vascular smooth phenotype. XXXVII Annual meeting of the chilean society of biochemistry and molecular biology. 2014.
- PDGF-BB induces mitochondrial degradation and autophagy in VSMCs. XXXVII Annual meeting of the chilean society of biochemistry and molecular biology. 2014.
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- Mitochondrial metabolism and the control of vascular smooth phenotype. XXXVII Annual meeting of the chilean society of biochemistry and molecular biology. 2014.
- PDGF-BB induces mitochondrial degradation and autophagy in VSMCs. XXXVII Annual meeting of the chilean society of biochemistry and molecular biology. 2014.

SCHOLAR **WEBSITE** **LINKEDIN**